Unit: 6 Unit 8: Quadratics **Name:** ______________
Topic: Converting Forms (Completing the Square) **Date:** ____________

Foundation (Jalapeno)

Q1. What is the vertex form of the equation $x^2 + 6x + 8$?
(A) $(x+3)^2 - 1$
(B) $(x+3)^2 + 1$
(C) $(x+4)^2 - 8$
(D) $(x+4)^2$
(E) $(x-3)^2 + 1$

Q2. Complete the square for $x^2 + 4x$
(A) $(x+2)^2 - 4$
(B) $(x+2)^2$
(C) $(x-2)^2 + 4$
(D) $(x+4)^2$
(E) $(x-4)^2$

Q3. What is the vertex of the parabola $y = x^2 + 2x + 1$?
(A) $(-1, 0)$
(B) $(-1, 1)$
(C) $(1, 0)$
(D) $(0, 1)$
(E) $(1, 1)$

Q4. Convert $y = x^2 - 4x - 3$ to vertex form
(A) $y = (x-2)^2 - 7$
(B) $y = (x+2)^2 - 7$
(C) $y = (x-2)^2 + 7$
(D) $y = (x+2)^2 + 7$
(E) $y = (x+1)^2 - 4$

Q5. Complete the square for $x^2 - 2x$
(A) $(x-1)^2 - 1$
(B) $(x+1)^2 - 1$
(C) $(x+1)^2$
(D) $(x-1)^2$
(E) $(x+2)^2 - 4$

Q6. What is the equation of the parabola with vertex $(-2, 3)$?
(A) $y = (x+2)^2 + 3$
(B) $y = (x-2)^2 + 3$
(C) $y = (x+2)^2 - 3$
(D) $y = x^2 - 4x + 3$
(E) $y = x^2 + 4x + 3$

Q7. Convert $y = x^2 + 2x - 6$ to vertex form
(A) $y = (x+1)^2 - 7$
(B) $y = (x-1)^2 - 7$
(C) $y = (x-1)^2 + 7$
(D) $y = (x+1)^2 + 7$
(E) $y = x^2 + 2x + 1$

Q8. Complete the square for $x^2 + 10x$
(A) $(x+5)^2 - 25$
(B) $(x-5)^2 - 25$
(C) $(x+5)^2 + 25$
(D) $(x+10)^2 - 100$
(E) $(x-10)^2 + 100$

Open Ended Questions (FRQ)

Q9. Convert $y = 2x^2 + 12x + 5$ to vertex form and identify the vertex

Q10. Complete the square for $x^2 - 8x + 3$ and write the equation in vertex form

Chef's Tips

- Tip 1: Ensure students show all work and justify their answers
- Tip 2: Encourage students to check their work by plugging the vertex form back into the standard form
- Tip 3: Remind students to identify the vertex from the vertex form of the equation